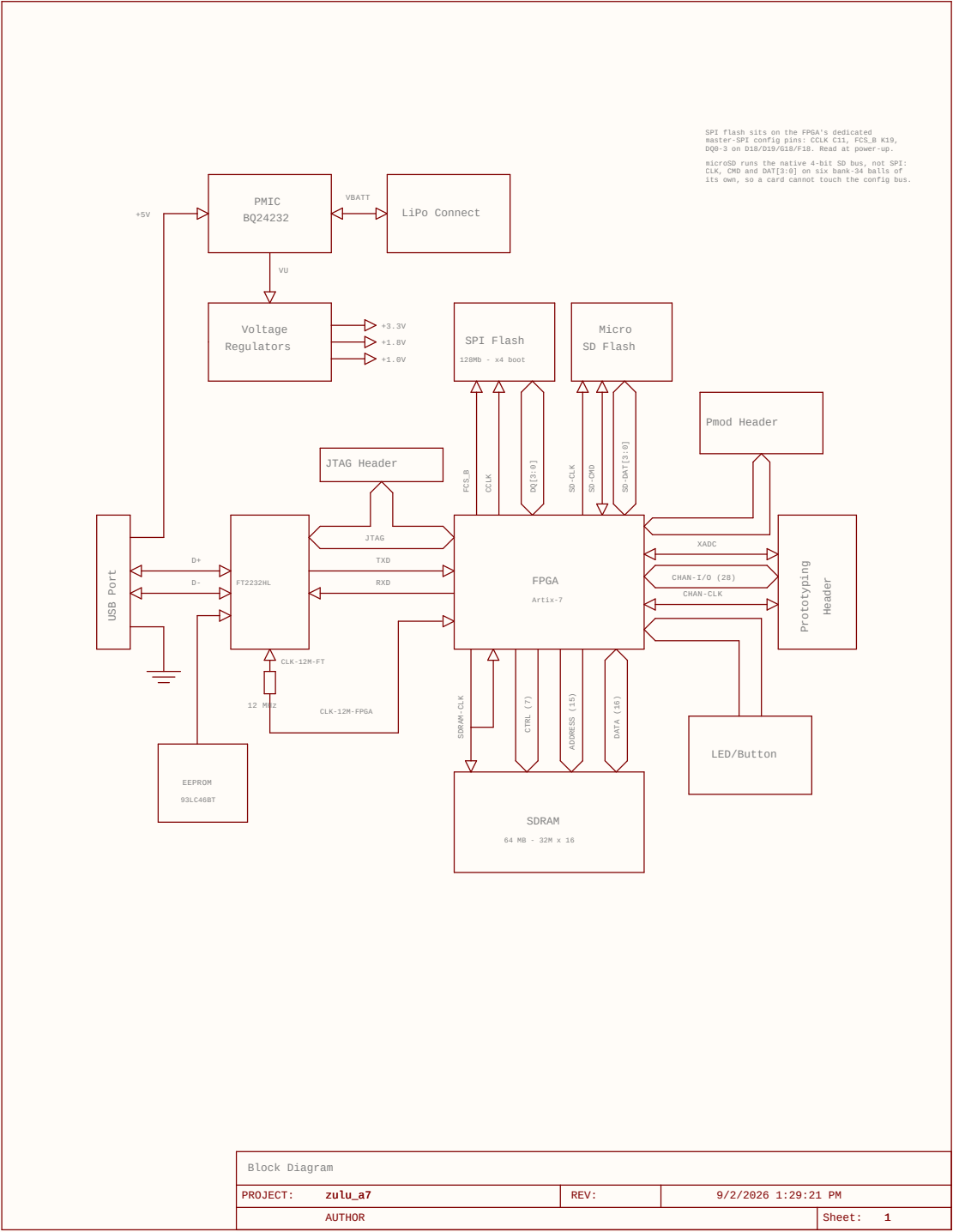
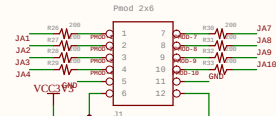
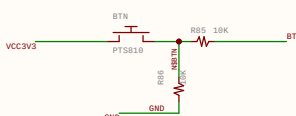
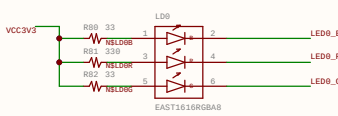
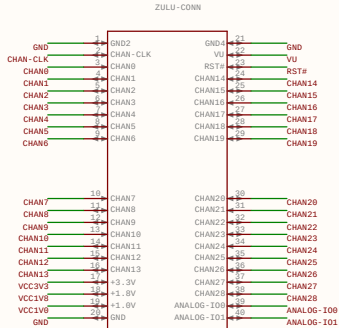




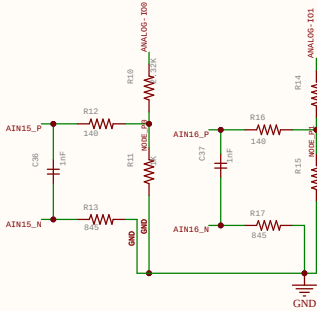


Power Supplies			
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X2: 40 pins on a 2.54 mm grid, rows 22.86 mm (0.900 in) apart, numbered 1-40 straight through from the right and skipping both landings. The two rows are mirror images. TOP ROW, x 50.69 down to 1.27: GND (pin 1), CHAN-CLK (2), CHAN0-CHAN6 (3-9), then CHAN7-CHAN13 (10-16), +3.3V, +1.8V, +1.0V, GND (17-20). BOTTOM ROW, x 50.69 down to 1.27: GND, VU, RST* (pins 21-23), CHAN14-CHAN16 (24-26), CHAN17-CHAN19 (27-29), then CHAN20-CHAN28 (30-38), ANALOG-IO0, ANALOG-IO1 (39, 40). FOUR positions in each row -- x 36.83, 34.29, 31.75, 29.21, between pins 9 and 10 above and between pins 29 and 30 below -- are RESERVED, not spare: the edge-mounted micro-USB X1 lands on the top row (7.88 mm of copper), the top-side L1Po header X4 (JST B2B-PH-SMA-1B, 7.95 mm body) lands on the bottom row directly opposite it.



Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the Pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmod A7 has none at its Pmod either. JA2-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16. JA7 is C15, also bank 16. JA8/JA9/JA10 are W16/W2/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.



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General IO			
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U3 = AS4C32M16SB-7TCN: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
AS4C16M16SA-6TIN it replaces -- all 54 pins and every
TSP: 1 dimension checked against both datasheets, and
pin 40 is NC on both.
One DIFFERENCE, and IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), not A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
-7TCN since 2026-09-08 (-6TIN dry): 143 MHz max at CL13,
tRCD/trp 21, tRC 63 ns, TA 0-70 C, IOD1 max 110 mA.

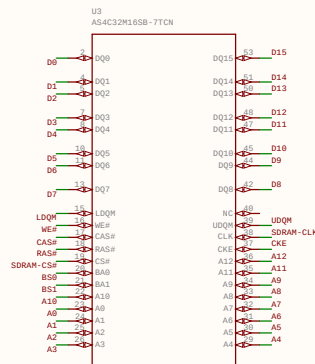


Figure 1: Schematic diagram of the proposed 8-bit DAC. The diagram shows a ladder network of 8-bit DAC cells. Each cell consists of a PMOS transistor (P1) and an NMOS transistor (N1) connected in series. The gates of P1 and N1 are connected to a common gate voltage (VDD/2). The sources of P1 and N1 are connected to a common source voltage (VSS). The drains of P1 and N1 are connected to a common drain voltage (VDD). The output of the DAC is taken from the node between the drains of P1 and N1. The schematic is labeled with VSS@3, GND, C3, VCC@3, VDD@3, C4, VCC@3, VDD@3, C5, VCC@3, VDD@2, C6, VCC@3, VDD@2, C7, VCC@3, VDD@1, C8, VCC@3, VDD@0, C9, VCC@3, VDD@0, and C10, VCC@3, VDD@0.

Memory			
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